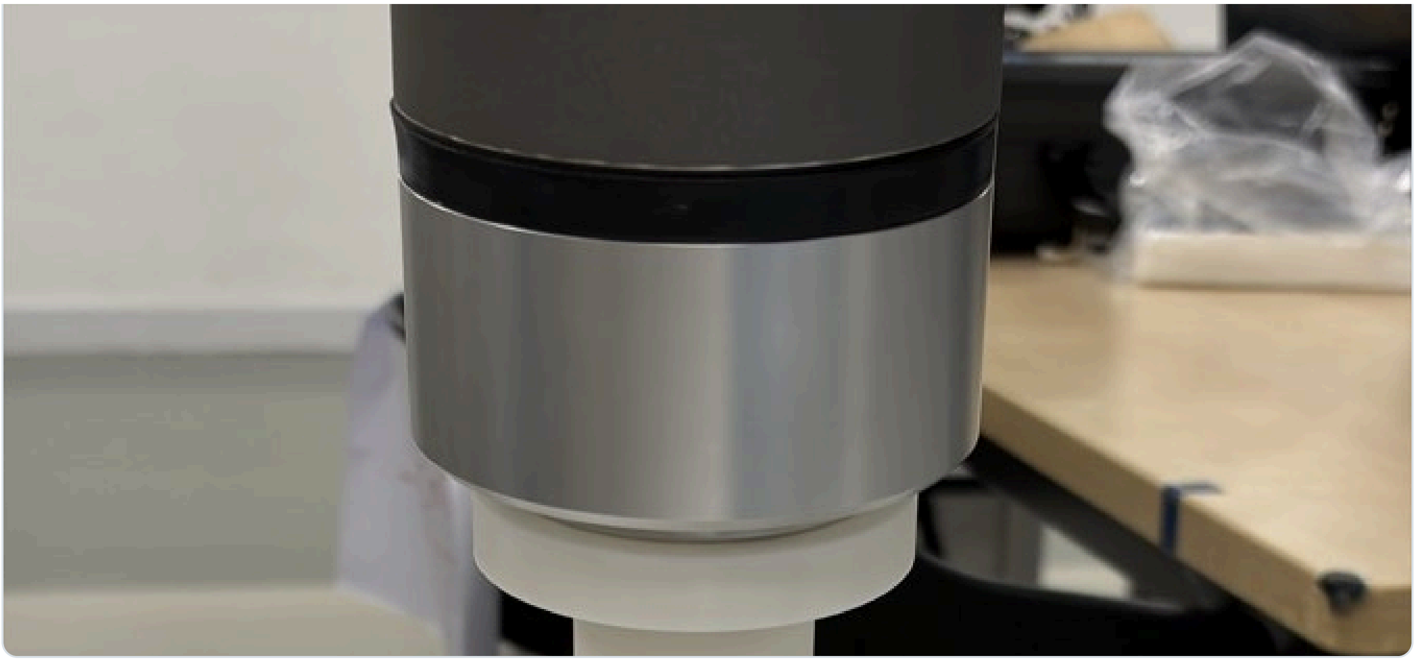
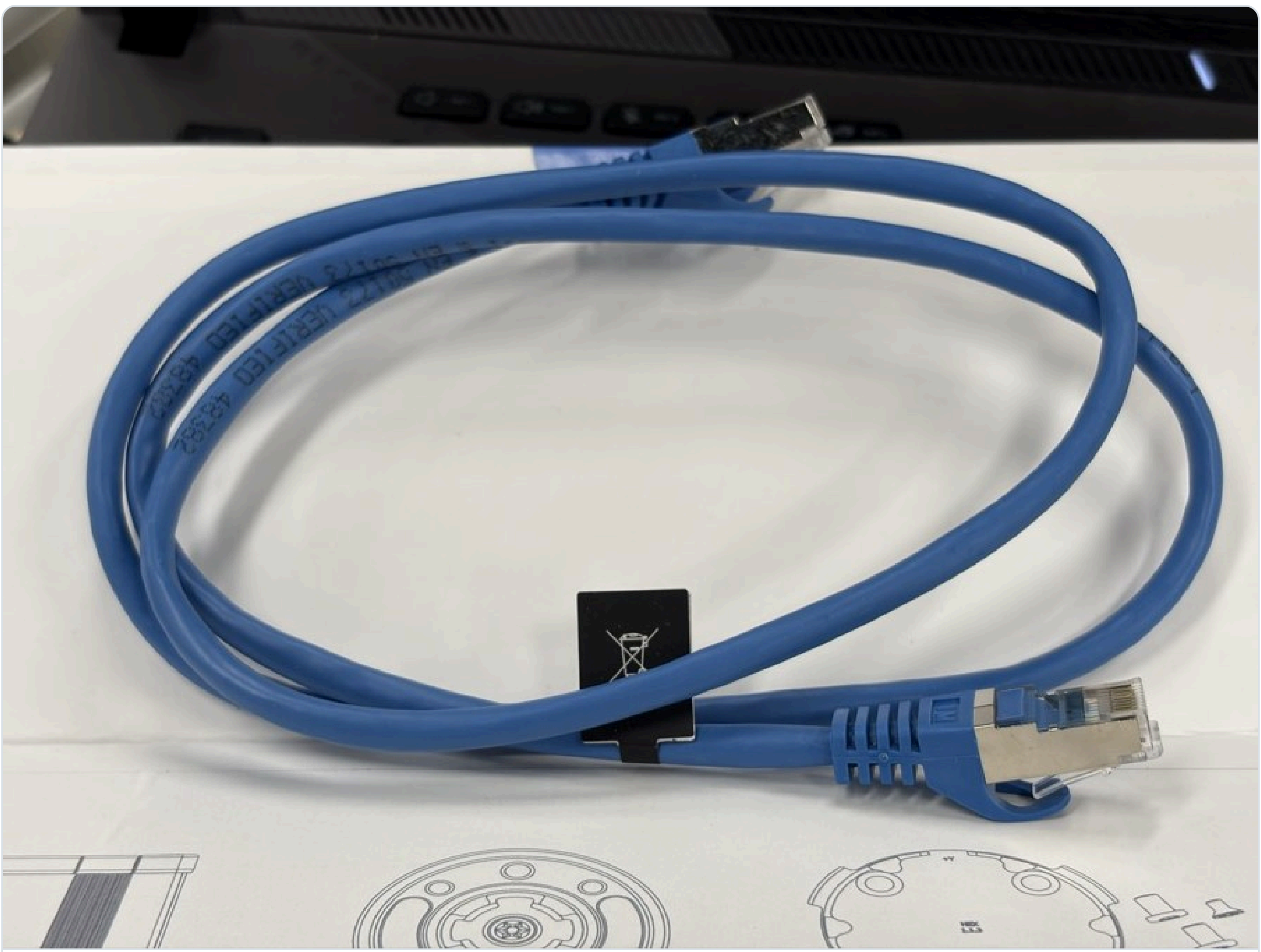
BENCH SETUP

UR10e / OnRobot HEX force-control bench

The meeting focus is the working experimental environment: the sensor is mounted, the contact surface is ready, and the Compute Box / Ethernet / RTDE path is wired for repeatable force-control measurements.


RATCH





Ethernet cable used in the Compute Box / switch path.

DATA PATH

Force data is now visible from the robot program into analysis plots

The bench uses the OnRobot URCap path for force-control commands and exports the measured wrench through RTDE output registers for Ubuntu-side logging.

HEX sensor

Tool-side force/torque measurement

↓

Compute Box

Device cable, power, fixed IP, Ethernet

↓

Switch + UR10e + Ubuntu

Shared network for URCap discovery and RTDE logging

↓

RTDE registers

Fx/Fy/Fz/Tx/Ty/Tz available in the run log

↓

Meeting plots

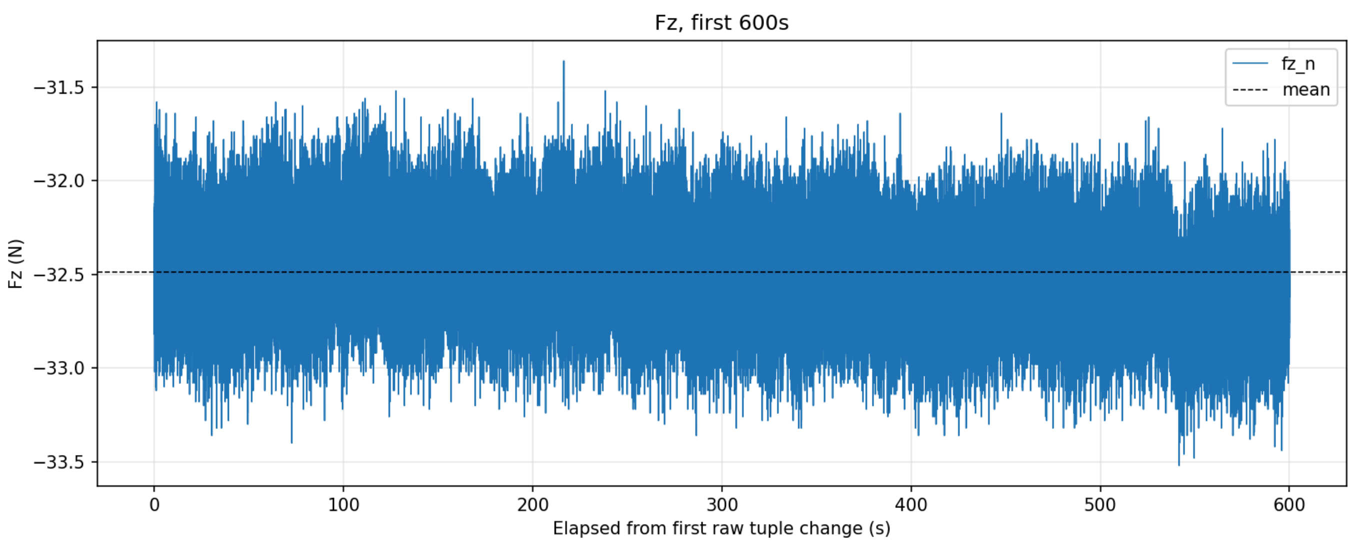
Force tracking and path tracking from the same run window

The final sensor comparison should be remeasured under one zeroing protocol. Existing plots below are placeholders for layout and story flow.

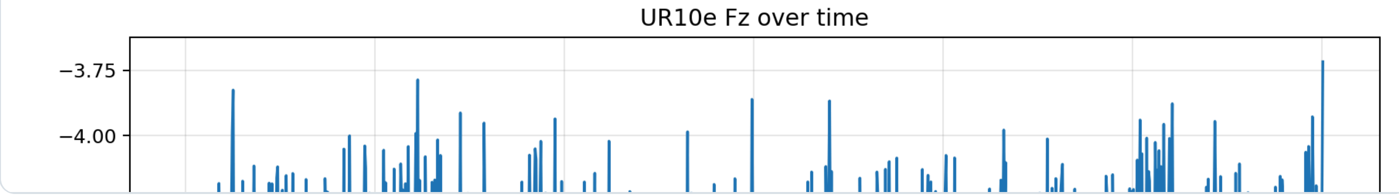
SENSOR BASELINE

Baseline slot for force-sensor validation

First draft uses existing data-path plots as placeholders. The final version should replace this with a zeroed 10-minute OnRobot run and a zeroed UR built-in F/T run under the same static condition.



OnRobot TCP DAQ Fz placeholder, 600 s. Not the final same-zero comparison.



FORCE-CONTROL DEMO

Closed-loop force tracking and contact path tracking

The demo result is evidence that the environment can run a force-controlled path. It is not the main story by itself.

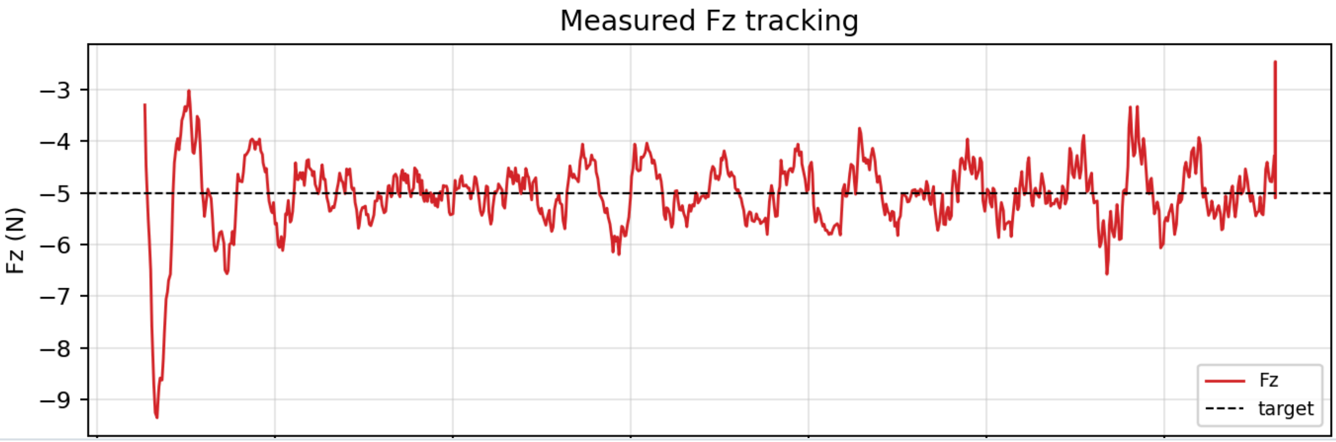
-5.080
Fz mean (N)

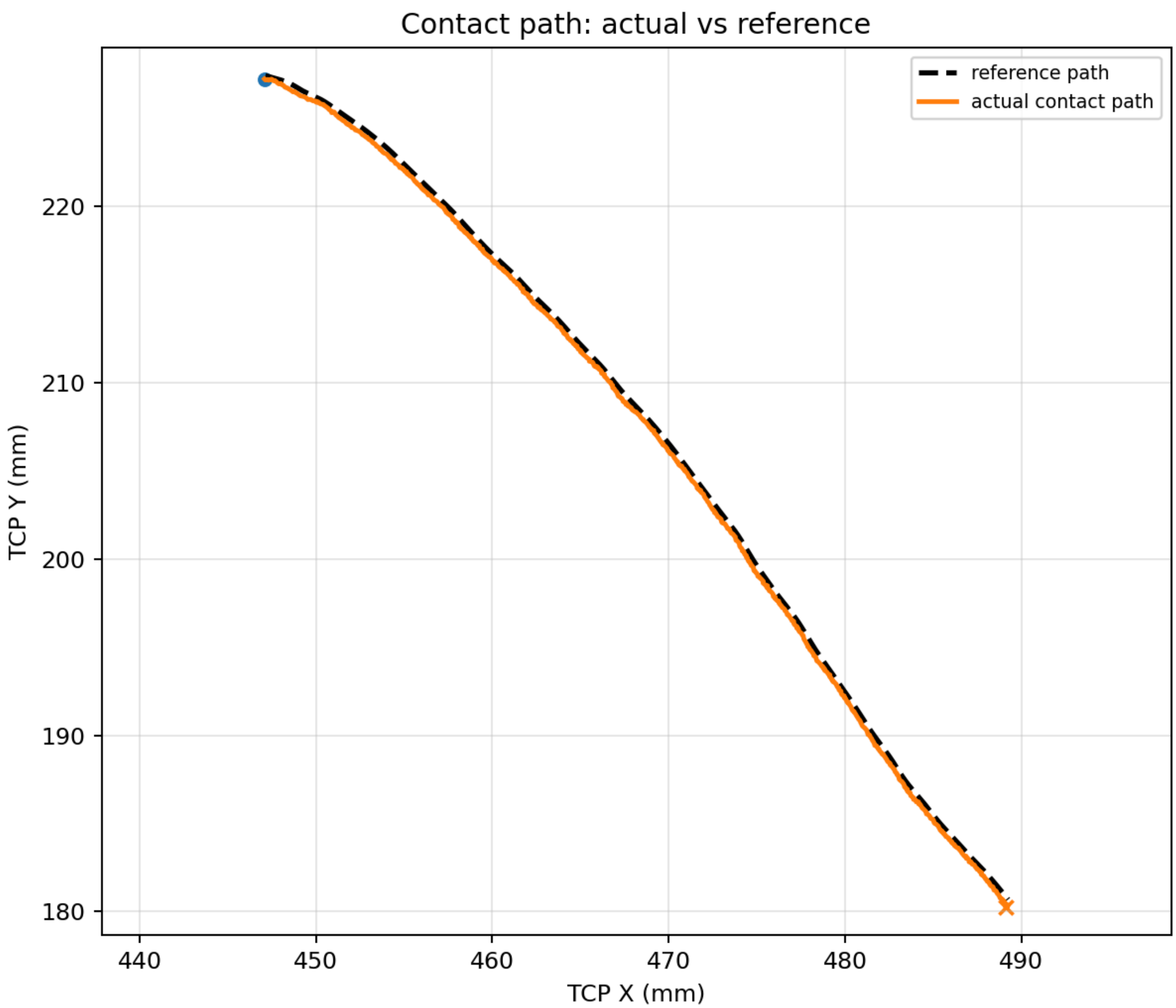
-0.080
Fz - target mean (N)

0.612
Fz error MAE (N)

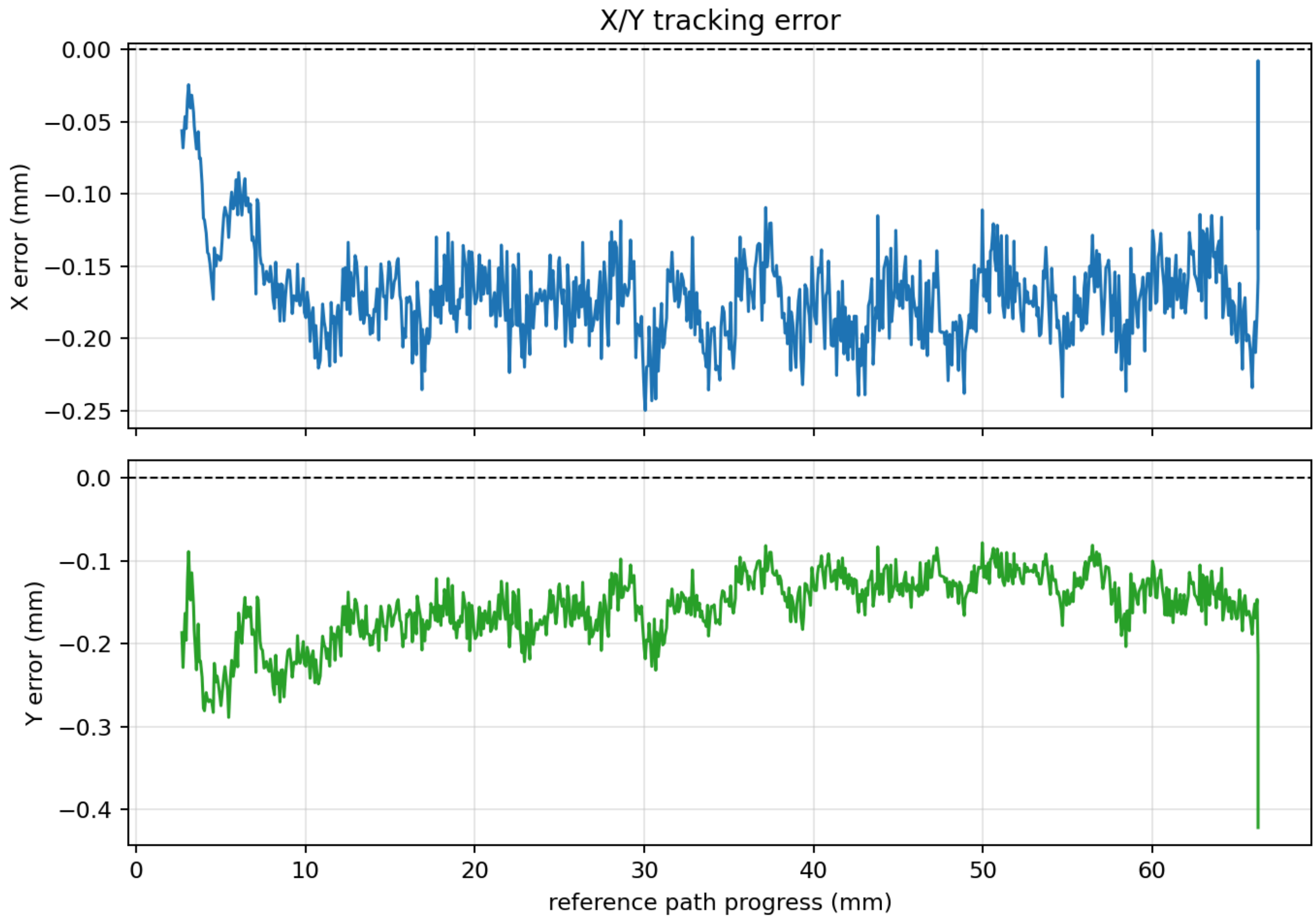
85.9%
within +/-1 N

0.304
2D path P95 (mm)





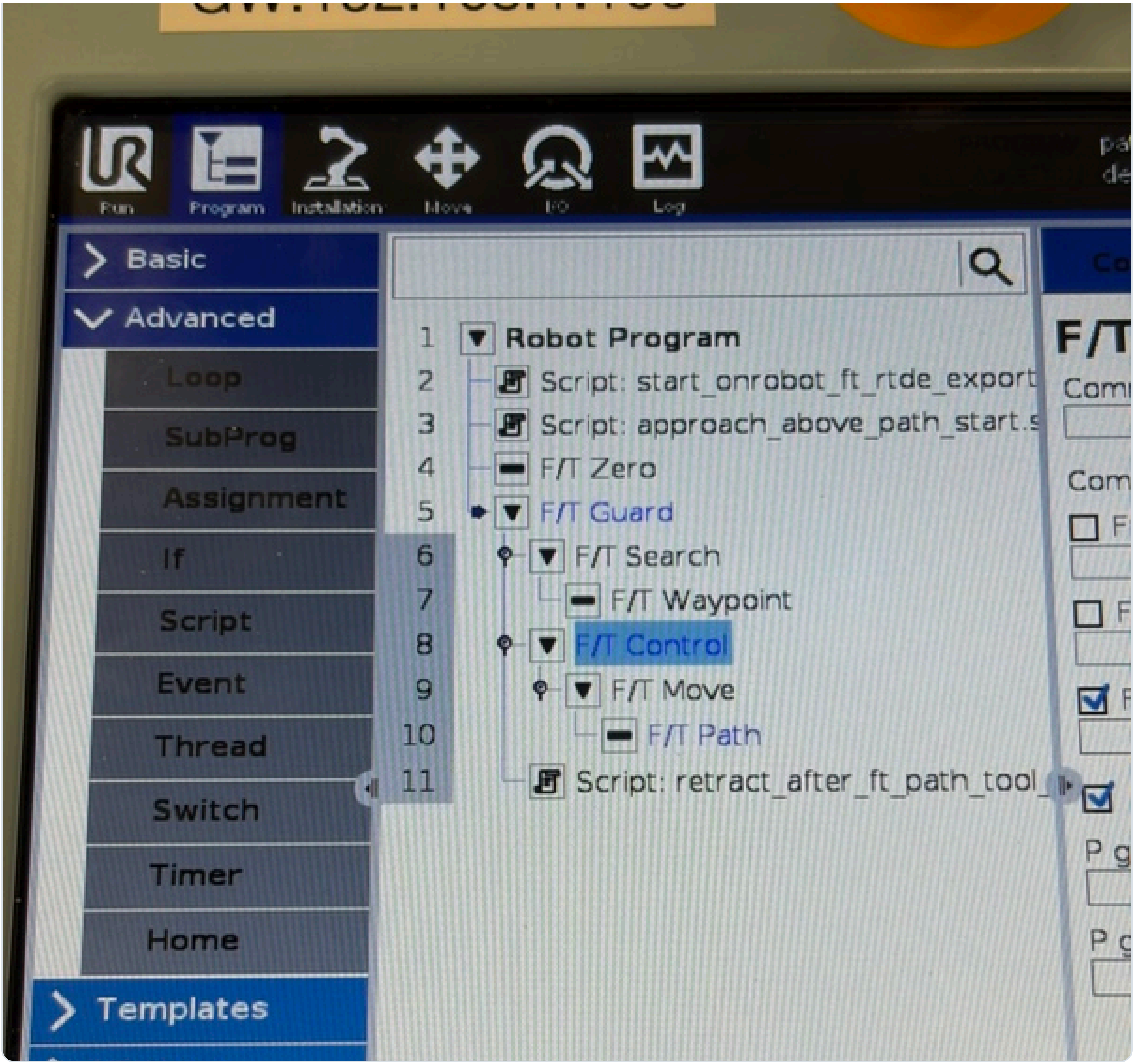
Actual contact path against reference path.



PROGRAM INTERFACE

URCap program structure

This clip is the PolyScope program interface, so it is named separately from the real robot demo video.



Program nodes
F/T Zero, F/T Search, F/T Control, F/T Move, F/T Path.

Settings kept in the meeting notes
Tool Z compliant, force PID, and F/T Move speed.

Demo video
Robot contact video pending; do not reuse the program-interface clip as the demo video.

NEXT VERSION

What to add before the meeting

Take or select clearer photos of the small Ethernet switch and the full Compute Box wiring.

Replace the baseline placeholders with a zeroed 10-minute OnRobot measurement.

Collect a same-zero UR built-in F/T comparison under the same static condition.

Add a true robot contact demo video, separate from the PolyScope program interface clip.

Optional: add a contact-surface 3D model or CAD view after the photos establish the physical setup.